

Sparse Bayesian modelling of underreported count data

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Abstract: We consider Bayesian inference for regression models of count data subject to underreporting. For the data generating process of counts as well as the fallible reporting process a joint model is specified, where the outcomes in both processes are related to a set of potential covariates. Identification of the joint model is achieved by additional information provided through validation data and incorporation of variable selection. For posterior inference we propose a convenient Markov chain Monte Carlo (MCMC) sampling scheme which relies on data augmentation and auxiliary mixture sampling techniques for this two-part model. Performance of the method is illustrated for simulated data and applied to analyse real data, collected to estimate risk of cervical cancer death.

Key words: MCMC, parameter identification, Poisson regression, underreporting, variable selection

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1 Introduction

Count data are collected in many fields like criminology, epidemiology or traffic safety to assess or monitor risks. However, often these data are subject to underreporting: Only a fraction of the true (but unobserved) counts is reported. Underreporting is a common problem, affecting count data collected in many applications. Examples where underreporting is discussed include cases of road crashes (Amoros *et al.*, 2006; Ma and Li, 2010), assaults (Moreno and Girón, 1998), victimization (Li *et al.*, 2003), cancer (Whittemore and Gong, 1991) and days absent from workplace (Winkelmann, 1996). More general misreporting of count data, where both under- and overreporting may occur, has also been considered, e.g., in Stamey *et al.* (2008), Sposto *et al.* (1992) and Bratcher and Stamey (2002).

Though information on unreported events is missing, underreporting is different from the concept of missing data as introduced in Rubin (1976), see also Little and Rubin (2002). There, information that data are missing is available and hence, can be incorporated in the analysis, whereas for an unreported event no information at all is generated.

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If underreporting occurs, inference from the observed counts will be biased, and consequently, risks will be underestimated if underreporting is not accounted for. To account for potential underreporting, specification of a joint model for the data generating process of events as well as the reporting process is required. A simple choice is to assume that the true counts are generated by a Poisson process and reporting/non-reporting is independent for each event. Specification of regression models allows us to relate both the intensity of the Poisson process as well as the reporting probability to a set of potential covariates. Combining a Poisson regression model with a logit model for underreporting yields the Poisson-Logistic (Pogit) model, proposed by Winkelmann and Zimmermann (1993). However, even in a simple model where both the intensity of the Poisson process as well as the reporting probability are assumed constant, the resulting model is not identified based solely on the reported count data (see, e.g., Papadopoulos and Santos Silva, 2012; Cameron and Trivedi, 2013). Identification requires additional information, which can be provided either by additional data on the reporting process, parameter restrictions or prior information on parameters, e.g., provided by experts. Moreno and Girón (1998) illustrate the incorporation of prior information in a Bayesian analysis of two data sets on assaults.

In this paper we will assume that in addition to a main study, which contains data on reported counts, a small set of validation data is available, which provides information on the reporting process for true cases. We perform a Bayesian analysis of the Pogit model and incorporate Bayesian variable selection. The main goal of Bayesian variable selection is to identify regressors with a non-zero effect; however, it provides an additional advantage in the Pogit model, as restrictions on the parameters in either the logit or the Poisson part of the joint model can be useful for identification of the remaining parameters (Papadopoulos and Santos Silva, 2012). Particularly, if the coefficient of a covariate is restricted to zero in one part, e.g., the logit model, the corresponding coefficient of the same covariate in the other part, the Poisson model, is strictly identified from only the main study data. Though identification is a minor issue with additional validation data and proper prior distributions, incorporation of variable selection can substantially reduce uncertainty on regression coefficients. To model underreported clustered data, we consider an extension of the Pogit model and include a cluster-specific random intercept in both the logit as well as the Poisson part of the joint model to account for within-cluster dependence and apply Bayesian variance selection to determine whether there is between-cluster variation in either part of the model.

For posterior inference we propose a convenient MCMC sampling scheme which relies on augmenting the observed data by the unobserved counts. As an advantage of this data augmentation step, the model for the augmented data can be factorized into a Poisson and a binomial logit part and hence, existing sampling schemes for both models can be used. We will use data augmentation as in Frühwirth-Schnatter *et al.* (2009) in the Poisson part and the algorithm of Fussl *et al.* (2013) for the binomial logit model. We additionally incorporate Bayesian variable selection in both parts of the model. For the logit model, an alternative to the data augmentation method in Fussl *et al.* (2013), which yields very accurate, but still approximations to the posterior distribution, is the method proposed recently by Polson *et al.* (2013), which is based

on the new family of Póly-Gamma distributions and allows to sample from the exact posterior.

We will use the Bayesian Pogit model to analyse data on cervical cancer deaths in four different European countries analysed previously in Whittemore and Gong (1991) and Powers *et al.* (2010). The main data set consists of the number of cervical cancer death cases in four countries and four age groups, as well as the number of woman-years at risk, and the validation study supplies information on the certification of a true cervical cancer case in the four countries. In contrast to previous analyses, where only main effects in the cancer death risk model were considered, we start from a saturated model with two-way interactions and determine the relevant effects using Bayesian variable selection.

The rest of the paper is structured as follows. Section 2 introduces the model and the prior distributions for all parameters. The MCMC scheme used for Bayesian inference is outlined in Section 3 and its performance is investigated for simulated data in Section 4. The application to the cervical cancer data is presented in Section 5 and finally, Section 6 summarizes the results and gives some concluding remarks.

2 Model specification

2.1 Joint regression modelling

We assume that the true, but unobserved (unreported) number of events n_i of unit i , $i = 1, \dots, I$, is generated by a Poisson process with intensity μ_i , i.e., $n_i \sim \mathcal{P}(\mu_i)$. The risk of an event in unit i is given as $\lambda_i = \mu_i/E_i$, where E_i denotes the total exposure time contributed by unit i . An event occurring for unit i is reported only with probability p_i and reporting of events is independent conditional on the reporting probability. Thus, the observed (reported) count data $\mathbf{y} = (y_1, \dots, y_I)$ are generated from the true number of events n_i by binomial thinning (Cameron and Trivedi, 2013), i.e., $y_i|n_i, p_i \sim \text{Bin}(n_i, p_i)$. As n_i is not known, inference has to be based on the observed counts y_i , which are Poisson distributed with intensity $\mu_i p_i$

$$y_i|p_i \sim \mathcal{P}(\mu_i p_i). \quad (2.1)$$

Obviously, if no further information is available, only the product $\eta_i = \mu_i p_i$ is identified from the observed data as any parameter combination with $\tilde{\mu}_i p_i = \eta_i$ yields the same likelihood. To achieve identification, additional information is required and we will return to this issue later.

We will assume that both the risk as well as the reporting probability in (2.1) depend on a set of covariates $\mathbf{x}_i$ and $\mathbf{w}_i$, respectively. The regression model for the log-risk is specified as

$$\log(\lambda_i) = \beta_0 + \mathbf{x}_i \boldsymbol{\beta}, \quad (2.2)$$

where β_0 is the intercept and $\boldsymbol{\beta}$ the $d_\beta \times 1$ vector of unknown regression effects. Obviously, equation (2.2) specifies a linear model for the log-intensity, if E_i is set to 1, $i = 1, \dots, I$.

For the reporting probabilities p_i we assume a logit regression model,

$$\log(p_i/(1 - p_i)) = \alpha_0 + \mathbf{w}_i\boldsymbol{\alpha}, \quad (2.3)$$

where α_0 is the intercept and $\boldsymbol{\alpha}$ the vector of regression coefficients of length d_α . The joint model given in equations (2.2) and (2.3) is a version of the Pogit model introduced in Winkelmann and Zimmermann (1993).

From a modelling point of view, $\mathbf{x}_i$ and $\mathbf{w}_i$ might be equal, overlapping or one might be a subset of the other. A convenient modelling strategy would be to include all available variables as regressors in both models. However, as shown in Papadopoulos and Santos Silva (2012), identification of the regression parameters $(\alpha_0, \boldsymbol{\alpha}, \beta_0, \boldsymbol{\beta})$ is not possible without additional information when the same regressors are used in both models. More specifically,

$$\begin{aligned} \lambda_i p_i &= \frac{\exp(\alpha_0 + \mathbf{x}_i\boldsymbol{\alpha})}{1 + \exp(\alpha_0 + \mathbf{x}_i\boldsymbol{\alpha})} \exp(\beta_0 + \mathbf{x}_i\boldsymbol{\beta}) = \\ &= \frac{\exp(-\alpha_0 - \mathbf{x}_i\boldsymbol{\alpha})}{1 + \exp(-\alpha_0 - \mathbf{x}_i\boldsymbol{\alpha})} \exp(\beta_0 + \alpha_0 + \mathbf{x}_i(\boldsymbol{\beta} + \boldsymbol{\alpha})), \end{aligned}$$

and hence, there are at least two sets of parameters yielding exactly the same likelihood. Identification of the model parameters can be achieved by a sign restriction or an exclusion restriction; see Papadopoulos and Santos Silva (2012) for details. Alternatively, identification is feasible by additional information, provided either through validation data or prior information, e.g., elicited from experts.

To illustrate identification of the model when additional information on the reporting process is available, we consider a model with only an intercept in both the Poisson as well as the logit part. Under a flat prior on both intercepts, β_0 and α_0 , the posterior distribution is constant, i.e., $p(\beta_0, \alpha_0|\mathbf{y}) = p(\beta^*, \alpha^*|\mathbf{y})$ for

$$\beta^* = \beta_0 + \alpha_0 - \alpha^* + \log((1 + e^{\alpha^*})/(1 + e^{\alpha_0})).$$

Figure 1 compares the posterior distribution for simulated data ($I = 1000$, $\beta_0^{\text{true}} = 1$, $\alpha_0^{\text{true}} = 1.8$) in three different scenarios: under a flat prior on α_0 when only counts are observed (left) and when information on the reporting process is available either through an informative prior (middle) or validation data (right). The maximum of the posterior (marked in red) is attained along a curve without additional information, but is well identified under an informative prior as well as with additional validation data. Further plots of the posterior under varying amount of information on the reporting process are shown in Section A of the electronic supplement to this paper.

Following Whittemore and Gong (1991) and Powers *et al.* (2010), we will assume that additional information is provided by a small sample of validation data, available for each $i = 1, \dots, I$, where m_i true cases are subject to the fallible reporting process and $v_i \leq m_i$ cases are reported. If the reporting process is the same as in the main sample, $v_i \sim \text{Bin}(m_i, p_i)$. We note here that Powers *et al.* (2010) use a slightly different

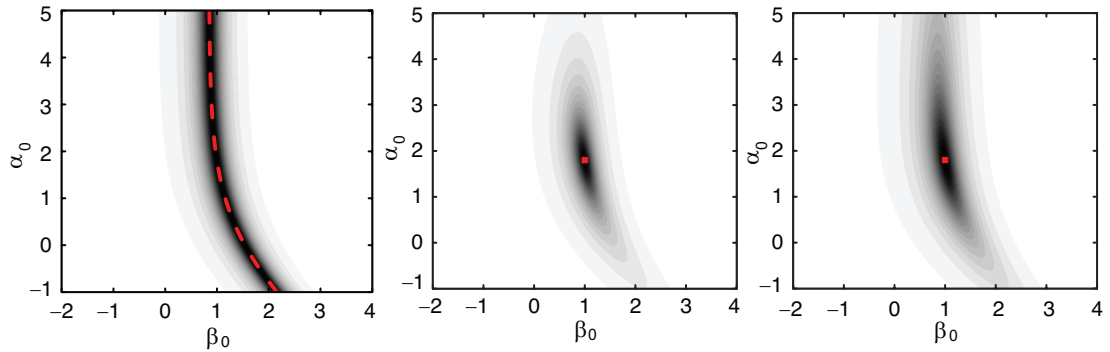


Figure 1 Poisson intercept model. Contour plots of the (scaled) log-posterior $\log p(\beta_0, \alpha_0 | \mathbf{y})^*$ for simulated data ($\beta_0^{\text{true}} = 1, \alpha_0^{\text{true}} = 1.8$) under a flat prior (left), under an informative prior (middle; $p(\alpha_0) \sim \mathcal{N}(\alpha_0^{\text{true}}, 0.01^2)$) and with validation data (right; 50 validation observations).

Source: Authors' own.

model for data of the same structure, as they additionally include a random intercept in the linear predictor of both regression models (2.2) and (2.3).

2.2 Modelling clustered data

As an extension we consider the case where count data are observed in clusters of units. To account for heterogeneity between and dependence within clusters, we include a cluster-specific random intercept in the linear predictor of the Poisson as well as the logit regression model

$$\log(p_i / (1 - p_i)) = \alpha_0 + \mathbf{w}_i \boldsymbol{\alpha} + a_{c(i)}, \quad (2.4)$$

$$\log(\lambda_i) = \beta_0 + \mathbf{x}_i \boldsymbol{\beta} + b_{c(i)}, \quad (2.5)$$

where $a_{c(i)} \sim \mathcal{N}(0, \sigma_a^2)$ and $b_{c(i)} \sim \mathcal{N}(0, \sigma_b^2)$ for each cluster $c = 1, \dots, C$. As we intend to perform selection of the random intercepts, we follow Tüchler (2008) and Wagner and Duller (2012) and perform posterior inference for the non-centred parameterizations of the models (2.4) and (2.5), given as

$$\log(p_i / (1 - p_i)) = \alpha_0 + \mathbf{w}_i \boldsymbol{\alpha} + \tilde{a}_{c(i)} \theta_\alpha, \quad \tilde{a}_{c(i)} \sim \mathcal{N}(0, 1), \quad (2.6)$$

$$\log(\lambda_i) = \beta_0 + \mathbf{x}_i \boldsymbol{\beta} + \tilde{b}_{c(i)} \theta_\beta, \quad \tilde{b}_{c(i)} \sim \mathcal{N}(0, 1). \quad (2.7)$$

Here, θ_α and θ_β are the signed standard deviations, $\theta_\alpha = \pm \sqrt{\sigma_a^2}$ and $\theta_\beta = \pm \sqrt{\sigma_b^2}$, which appear in the model equations (2.6) and (2.7) as regression effects, whereas the standardized random effects play the role of covariates. Hence, if say $\theta_\alpha = 0$, all random intercepts $a_{c(i)} = \theta_\alpha \tilde{a}_{c(i)}$ will be zero and the random intercept can be excluded from the linear predictor. On the other hand, if $|\theta_\alpha|$ is large, the random intercepts $a_{c(i)}$ will take large values and should be included in the model.

2.3 Prior distributions

Bayesian model specification is completed by assigning prior distributions to all model parameters. We assume prior independence of the parameters in the logit and the Poisson model, and hence, the prior for the parameters of the random intercept model is given as

$$p(\alpha_0, \boldsymbol{\alpha}, \theta_\alpha, \beta_0, \boldsymbol{\beta}, \theta_\beta) = p(\alpha_0, \boldsymbol{\alpha}, \theta_\alpha)p(\beta_0, \boldsymbol{\beta}, \theta_\beta).$$

The prior has the same structure for the parameters of both models, e.g., $p(\alpha_0, \boldsymbol{\alpha}, \theta_\alpha) = p(\alpha_0)p(\boldsymbol{\alpha})p(\theta_\alpha)$ in the logit model. If no random intercept is included, the corresponding prior of the random intercept parameter θ is omitted.

We assign flat but proper normal priors to the intercepts α_0 and β_0 , $\alpha_0 \sim \mathcal{N}(m_{\alpha,0}, M_{\alpha,0})$ and $\beta_0 \sim \mathcal{N}(m_{\beta,0}, M_{\beta,0})$. The regression effects are commonly assigned normal priors. However, as we intend to perform variable selection, we will specify mixture priors with a spike and slab component (see, e.g., Mitchell and Beauchamp, 1988; George and McCulloch, 1993, 1997). The spike component is concentrated at zero to allow shrinkage of small effects to zero, whereas the slab is rather flat, often with heavier tails, to prevent shrinkage of large effects.

To specify the priors hierarchically, we introduce for each component of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ an indicator variable $\delta_{\alpha,j}$ and $\delta_{\beta,j}$ for the slab component and assume conditional prior independence of the regression effects. The joint prior of regression effects $\boldsymbol{\beta}$ and indicators $\boldsymbol{\delta}_\beta$ is then given as

$$p(\boldsymbol{\beta}, \boldsymbol{\delta}_\beta) = \prod_{j=1}^{d_\beta} p(\beta_j | \delta_{\beta,j}) p(\delta_{\beta,j}), \quad (2.8)$$

where

$$p(\delta_{\beta,j} = 1 | \omega_\beta) = \omega_\beta, \quad (2.9)$$

$$p(\beta_j | \delta_{\beta,j}) = (1 - \delta_{\beta,j}) p_{\text{spike}}(\beta_j) + \delta_{\beta,j} p_{\text{slab}}(\beta_j). \quad (2.10)$$

This specification is particularly useful, as the posterior inclusion probability, i.e., the posterior probability of $\{\delta_{\beta,j} = 1\}$, allows to classify regression effects as zero or non-zero.

We will use a Dirac spike, i.e., a point mass at zero, $p_{\text{spike}}(\beta_j) = \Delta_0(\beta_j)$, $j = 1, \dots, d_\beta$ and a Student- t slab. More specifically, the slab component $p_{\text{slab}}(\beta_j)$ is the pdf of the $t_{2\nu}(0, Q/\nu)$ -distribution, which is specified as a scale mixture of normal distributions,

$$\beta_j | (\delta_{\beta,j} = 1, \psi_{\beta,j}) \sim \mathcal{N}(0, \psi_{\beta,j}), \quad (2.11)$$

where $\psi_{\beta,j} \sim \mathcal{G}^{-1}(\nu, Q)$. Finally, we define a hyper-prior for the mixture weight ω_β , $\omega_\beta \sim \mathcal{B}(a_{\beta,0}, b_{\beta,0})$. We assign a prior of exactly the same structure to $\boldsymbol{\alpha}$. Formally, the prior is obtained by replacing subscripts β in equations (2.8)–(2.11) by α .

In the random intercept model given in equations (2.6) and (2.7), priors have to be specified also for the parameters θ_α and θ_β . As these parameters can be interpreted as regression effects of the auxiliary covariates, defined by the standardized random intercepts $\tilde{a}_{c(i)}$ and $\tilde{b}_{c(i)}$, priors can be specified analogously to those for α and β . More specifically, we define the indicator γ_β in the Poisson model and specify the prior for θ_β as follows

$$\begin{aligned} p(\gamma_\beta = 1 | \pi_\beta) &= \pi_\beta, \\ p(\theta_\beta | \gamma_\beta) &= (1 - \gamma_\beta) \mathbf{p}_{\text{spike}}(\theta_\beta) + \gamma_\beta \mathbf{p}_{\text{slab}}(\theta_\beta), \end{aligned}$$

where the slab component has a $\mathcal{N}(0, \psi_{\beta, d_\beta+1})$ -distribution and the variance parameter $\psi_{\beta, d_\beta+1}$ also is assigned an inverse Gamma distribution, $\psi_{\beta, d_\beta+1} \sim \mathcal{G}^{-1}(\nu, Q)$. Again, the hierarchical prior of θ_α is specified in the same way and is obtained by replacing the subscript β by α .

We will denote by Θ_β the collection of all parameters and hyper-parameters of the Poisson model, i.e., $\Theta_\beta = (\beta_0, \boldsymbol{\beta}, \boldsymbol{\delta}_\beta, \boldsymbol{\psi}_\beta, \omega_\beta)$ in the regression model, and Θ_β additionally includes $(\theta_\beta, \gamma_\beta, \psi_{\beta, d_\beta+1}, \pi_\beta)$ in the random intercept model. Correspondingly, Θ_α denotes the parameters of the logit model.

3 Bayesian inference

Bayesian inference is performed by sampling from the posterior using MCMC methods and data augmentation. To simplify the structure of the likelihood and to facilitate posterior inference, we augment the data with the unobserved counts $d_i = n_i - y_i$. We present the basic sampling scheme in Section 3.1 and give details on posterior sampling of the parameters of the logit and the Poisson model in Sections 3.2 and 3.3.

3.1 Posterior sampling scheme

The unobserved counts $d_i = n_i - y_i$ follow a Poisson distribution

$$d_i \sim \mathcal{P}\left(E_i \lambda_i(\Theta_\beta)(1 - p_i(\Theta_\alpha))\right).$$

The complete data likelihood of observed and augmented data, $(\mathbf{y}, \mathbf{v})$ and $\mathbf{d} = (d_1, \dots, d_1)$ respectively, is then given as

$$\begin{aligned} p(\mathbf{y}, \mathbf{v}, \mathbf{d} | \Theta_\alpha, \Theta_\beta) &= \prod_{i=1}^I p(y_i, d_i | \lambda_i(\Theta_\beta), p_i(\Theta_\alpha)) \prod_{i=1}^I p(v_i | p_i(\Theta_\alpha)) \\ &\propto \prod_{i=1}^I (E_i \lambda_i(\Theta_\beta))^{y_i+d_i} e^{-E_i \lambda_i(\Theta_\beta)} \prod_{i=1}^I p_i(\Theta_\alpha)^{y_i+v_i} (1 - p_i(\Theta_\alpha))^{m_i-v_i+d_i} \end{aligned} \quad (3.1)$$

The contribution of unit i to the complete data likelihood in equation (3.1) thus factorizes in the Poisson likelihood for $n_i = y_i + d_i$, which depends only on Θ_β , and the binomial likelihood for $y_i + v_i$, involving only Θ_α . The augmented posterior is therefore given as

$$p(\Theta_\alpha, \Theta_\beta | \mathbf{y}, \mathbf{v}, \mathbf{d}) \propto p(\Theta_\alpha) \prod_{i=1}^I p(y_i + v_i | d_i, \Theta_\alpha) \cdot p(\Theta_\beta) \prod_{i=1}^I p(y_i + d_i | \Theta_\beta),$$

which suggests the following scheme for posterior sampling:

1. Sample the number of unreported cases $d_i \sim \mathcal{P}(E_i \lambda_i (1 - p_i))$ and compute the total number of cases $n_i = y_i + d_i$.
2. Sample the parameters Θ_α from the posterior of the binomial logit model for $\tilde{y}_i = y_i + v_i$ reported and $n_i + m_i - \tilde{y}_i = \tilde{n}_i - \tilde{y}_i$ unreported cases.
3. Sample the parameters Θ_β from the posterior of the Poisson regression model for the augmented data n_i .

In the regression models specified in Section 2.1, variable selection is performed in steps 2 and 3 by sampling the indicator variables δ_α and δ_β . As discussed in Wagner and Duller (2012), specifying a Dirac spike has consequences on posterior sampling: To guarantee irreducibility of the Markov chain, the indicator variables have to be sampled marginalizing over the corresponding regression effects α and β . This is computationally feasible for linear Gaussian regression models under conjugate normal priors, where the marginal likelihoods are available analytically. Representations as Gaussian regression models in auxiliary variables have been discussed in the literature, e.g., in Holmes and Held (2006), Tüchler (2008), Fussl *et al.* (2013) and Polson *et al.* (2013) for the logit and in Frühwirth-Schnatter *et al.* (2009) for the Poisson model. We will use representations, where the error term in the auxiliary model is a finite mixture of normal distributions and give details in Sections 3.2 and 3.3. Variable selection in normal regression models is summarized in Appendix A. The MCMC sampling scheme is implemented in the R package `pogit` (Dvorzak and Wagner, 2015), which is available on Comprehensive R Archive Network (CRAN). Example scripts and data sets are provided inside the R package and also as electronic supplements of the paper.

If the information on all or some parameters of the reporting process is not provided by validation data but by information encoded in a prior distribution for (α_0, α) , slight modifications are necessary only in sampling step 2 and will be discussed in Section 3.2. Similarly, information on the parameters of the Poisson sub-model affects only sampling step 3; see Section 3.3 for details.

Finally, we note that our sampling scheme can be interpreted as an alternative to the method presented in Powers *et al.* (2010) for Bayesian variable selection in regression models of underreported count data. As mentioned earlier, Powers *et al.* (2010) additionally include a unit-specific, normal random intercept in both the logit

and the Poisson regression model, and hence, the linear predictor in the Poisson model is given as

$$\log(\lambda_i) = \beta_0 + \mathbf{x}_i\boldsymbol{\beta} + b_i.$$

To perform variable selection, spike and slab prior distributions with Dirac spikes and slightly different, so-called g-type, slabs are specified for the regression coefficients subject to selection; see Powers *et al.* (2010) for details. Inclusion of the random intercept introduces variability in the linear predictor and allows to sample the indicators from the linear predictor, when marginalizing over the regression effects subject to selection which – as noted earlier – is essential to avoid reducibility of the Markov chain. The resulting MCMC scheme however does not sample from the posterior of the Pogit model, but from an over-dispersed Pogit model.

3.2 Data augmentation for binomial logit models

Data augmentation for the binomial logit model in Frühwirth-Schnatter *et al.* (2009) is based on the interpretation of the likelihood contribution of unit i as resulting from $\tilde{n}_i$ independent binary observations with $\tilde{y}_i$ ones and $\tilde{n}_i - \tilde{y}_i$ zeros. However, with this strategy the number of auxiliary observations would be $\sum \tilde{n}_i$, which could be enormous for large observed counts, e.g., count data in register systems. Therefore, we will follow Fussl *et al.* (2013), who show that the binomial logit model given in equation (2.6) can be represented as a linear regression model in the latent variable y_i^* , which has an interpretation as the difference of aggregated utilities as

$$y_i^* = \alpha_0 + \mathbf{w}_i\boldsymbol{\alpha} + \tilde{a}_{c(i)}\theta_\alpha + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{LG}(\tilde{n}_i).$$

Here, $\mathcal{LG}(\tilde{n}_i)$ denotes the Type III generalized logistic distribution with parameter $\tilde{n}_i$. Fussl *et al.* (2013) approximate the error distribution by a finite scale mixture of normal distributions to achieve an auxiliary Gaussian regression model

$$y_i^* = \alpha_0 + \mathbf{w}_i\boldsymbol{\alpha} + \tilde{a}_{c(i)}\theta_\alpha + \varepsilon_{r_i}, \quad \varepsilon_{r_i} \sim \mathcal{N}(0, s_{r_i}^2), \quad (3.2)$$

where r_i denotes the mixture component. For sampling details of the auxiliary variables y_i^* and r_i , we refer to Fussl *et al.* (2013). Equation (3.2) defines the observation equation of a linear Gaussian regression model with heteroscedastic errors and hence, step 2 is completed with variable selection and parameter estimation as detailed in Appendix A. Aggregation leads to a considerable reduction of computing time compared to augmenting the data with $\tilde{n}_i$ latent utilities as in Tüchler (2008), particularly if $\tilde{n}_i$ is large.

If information on the parameters $(\alpha_0, \boldsymbol{\alpha})$ of the reporting process is available, a straightforward strategy is to encode this in an informative normal prior distribution, $\mathcal{N}(\mathbf{a}_0, \mathbf{A}_0)$, instead of the spike and slab prior. As conditioning on the auxiliary variables, $(\alpha_0, \boldsymbol{\alpha})$ are regression coefficients in a standard Bayesian regression model with heteroscedastic errors (see equation (3.2)), the conditional posterior again is a normal distribution and can be sampled easily.

3.3 Auxiliary mixture sampling for Poisson models

To achieve a representation of the Poisson model as a Gaussian regression model, we perform data augmentation as in Frühwirth-Schnatter *et al.* (2009). We interpret n_i as the number of jumps in the unit time interval of an unobserved Poisson process with intensity $E_i \lambda_i(\Theta_\beta)$ and introduce the latent variables τ_i^* , where $\tau_i^* = (\tau_{i,1}^*, \tau_{i,2}^*)$, if $n_i > 0$ and $\tau_i^* = \tau_{i,1}^*$, if $n_i = 0$. $\tau_{i,1}^*$ is the interarrival time of the first jump after the unit interval, whereas $\tau_{i,2}^*$ is the arrival time of the n_i -th event in the unit interval. The Poisson regression model with risk specified in equation (2.7) is then equivalent to the linear regression model

$$-\log \tau_{i,k}^* - \log E_i = \beta_0 + \mathbf{x}_i \boldsymbol{\beta} + \tilde{b}_{c(i)} \theta_\beta + \varepsilon_{i,k},$$

with $k = 1, 2$, where $\varepsilon_{i,1}$ follows a Type I extreme value distribution and $\varepsilon_{i,2}$ has a negative log-Gamma distribution, i.e., $\varepsilon_{i,2} = -\log \xi_{i,2}$ with $\xi_{i,2} \sim \mathcal{G}(n_i, 1)$. Approximating the error distribution again by a finite mixture of normals yields the auxiliary linear Gaussian regression model

$$-\log \tau_{i,k}^* - \log E_i = \beta_0 + \mathbf{x}_i \boldsymbol{\beta} + \tilde{b}_{c(i)} \theta_\beta + \varepsilon_{i,k}^*, \quad (3.3)$$

with $\varepsilon_{i,k}^* | r_{i,k} \sim \mathcal{N}(m_{r_{i,k}}, s_{r_{i,k}}^2)$. Variable selection and parameter estimation in model (3.3) is performed as described in Appendix A. For details on data augmentation, see Frühwirth-Schnatter *et al.* (2009).

Just as in the logit model (see Section 3.2), prior information on the parameters $(\beta_0, \boldsymbol{\beta})$ encoded in an informative normal prior distribution, $\mathcal{N}(\mathbf{b}_0, \mathbf{B}_0)$, allows us to sample the parameters from the posterior distribution of a linear Gaussian regression model with heteroscedastic errors (see equation (3.3)).

4 Simulated data

In this section, we investigate the performance of the MCMC method for independent and clustered observations. In the first simulation study, we consider independent observations with four binary variables as potential regressors in both the Poisson and the logit model. In the second simulation study, the linear predictor in both models additionally includes a random intercept to model within-cluster dependence.

In both simulations, we specify normal priors for the intercepts with $m_{\alpha,0} = m_{\beta,0} = 0$ and $M_{\alpha,0} = M_{\beta,0} = 100$. Following Wagner and Duller (2012), we set the hyper-parameters ν and Q of slab variances to $\nu=5$ and $Q=20$, resulting in Student- t slabs with $2\nu = 10$ degrees of freedom. Finally, for the component weights ω_α , ω_β , π_α and π_β , we specify a uniform prior distribution, $\mathcal{B}(1, 1)$.

The choice for the spike and slab prior corresponds to a marginal prior probability of 0.83 that a regression effect β_j or α_j is in the interval $[-2, 2]$. Results of a

sensitivity study reported in Section B of the electronic supplement, where we investigate different values for the scale parameter Q , indicate that this is a sensible choice.

Posterior inclusion probabilities of the regression effects β are estimated from the MCMC output by

$$\hat{p}_{\beta,j} = \hat{p}(\delta_{\beta,j} = 1|\mathbf{y}) = \frac{1}{M} \sum_{m=1}^M p^{(m)}(\delta_{\beta,j} = 1|\mathbf{y}, \sim),$$

for $j = 1, \dots, d_\beta$ and correspondingly for α as well as for the random intercept parameters θ_α and θ_β .

4.1 Simulation I

We generated 300 data sets for $I = 16$ units. The same variables are used in both sub-models, i.e., $\mathbf{x}_i = \mathbf{w}_i$, and one observation is generated for each covariate pattern. The regression effects are set to $\beta_0 = 0.75$ and $\beta = (0.5, -2, 0, 0)$ in the Poisson and to $\alpha_0 = 2.2$ and $\alpha = (-1.9, 0, 0, 0)$ in the logit model. Observed data y_i were generated from a Poisson distribution with intensity $\eta_i = E_i \lambda_i p_i$, where $E_i = 100$ for each category, and validation data were generated from a $\text{Bin}(m_i, p_i)$ -distribution with sizes $m_i = 25, i = 1, \dots, 16$. We note here that compared to common epidemiological or public health studies, counts are rather small. The set-up is almost the same as that of the first simulation study in Powers *et al.* (2010); however, we include no random intercept. For each data set, we ran MCMC for $M = 8000$ iterations after a burn-in period of 4000 draws. The first 2000 iterations of the burn-in period were drawn from the unrestricted model including all regressors.

Table 1 summarizes the results. We report the average of the posterior means of the regression parameters over 300 data sets, the average of the posterior inclusion probabilities $\hat{p}_{\beta,j}$ and $\hat{p}_{\alpha,j}$, the nominal coverage of 95%-HPD intervals as well as the

Table 1 Simulation I. Averaged posterior means (Av. Pm.), averaged estimated posterior inclusion probabilities (Av. $\hat{p}_{*,j}$), nominal coverage of (model averaged) 95%-HPD intervals (Cov.) as well as number of models (N_{mod}) where a regressor is included in the final model over 300 data sets. Results for non-zero effects are given in bold.

Model	Covariate	True value	Av. Pm.	Av. $\hat{p}_{*,j}$	Cov.	N_{mod}
Logit	α_0	2.2	2.237	—	0.95	—
	α_1	−1.9	−1.922	1.000	0.94	300
	α_2	0	−0.010	0.140	1.00	8
	α_3	0	0.003	0.103	1.00	4
	α_4	0	0.005	0.102	1.00	3
Poisson	β_0	0.75	0.750	—	0.92	—
	β_1	0.5	0.496	0.999	0.96	300
	β_2	−2.0	−2.000	1.000	0.97	300
	β_3	0	0.000	0.077	1.00	1
	β_4	0	−0.001	0.076	1.00	4

Source: Authors' own.

number of models where the respective regressor is included in the final model, i.e., where $\hat{p}_{\beta,j} > 0.5$ and $\hat{p}_{\alpha,j} > 0.5$, respectively.

Regression effects are perfectly classified as zero or different from zero in the Poisson regression model with averaged posterior inclusion probabilities close to zero and one, respectively. Classification is less perfect in the logit model, where the average posterior inclusion probability is higher for zero effects. Based on posterior inclusion probabilities above 0.5, in 15 data sets one zero effect is misclassified as non-zero in the logit part of the joint model. Sensitivity of variable selection with respect to the slab variance of the prior distribution is investigated in Section B.1 of the supplement.

Figure 2 shows the variation of posterior means of the risks and the reporting probabilities for the 16 covariate patterns in the 300 simulated data sets (displayed by the grey lines) as well as the averages over all data sets (displayed by the dashed lines with crosses) and the true values (displayed by the solid lines with circles). Posterior means vary considerably across the 300 data sets, but their averages agree very well with the true values.

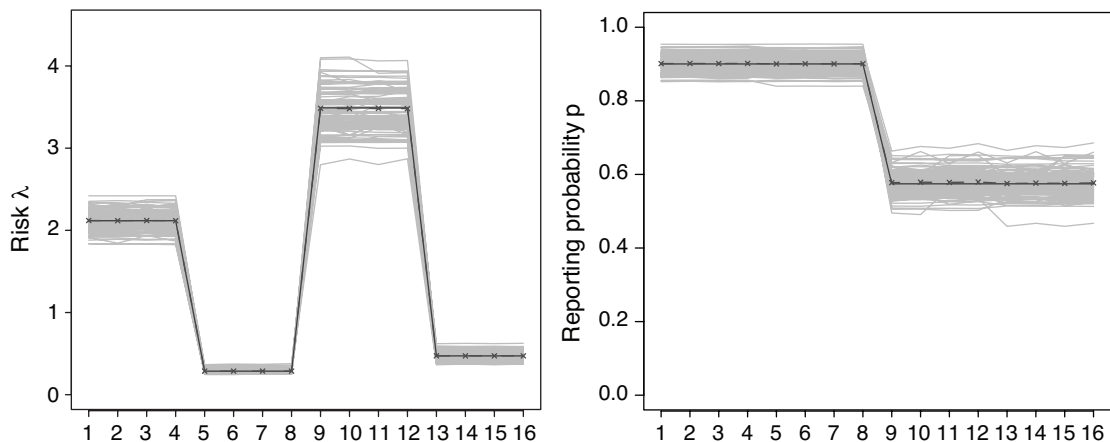


Figure 2 Simulation I. Posterior means (grey), averaged posterior means (crosses) and true values (circles) for risks λ_i and reporting probabilities p_i over 300 data sets for the 16 covariate patterns.

Source: Authors' own.

4.2 Simulation II

In the second simulation study, we generated 50 data sets for observations in $C = 50$ clusters. Again we used four binary covariates and built clusters containing one unit with each of the 16 covariate patterns, i.e., a total of $I = 800$ units. Regression effects were set to $\beta_0 = 0.75$ and $\beta = (0.1, 0.1, 0, 0)$ in the Poisson and to $\alpha_0 = 2.2$ and $\alpha = (-0.3, 0, -0.3, 0)$ in the logit model. Random intercepts a_c and b_c in the logit and the Poisson model, respectively, were drawn independently from the $\mathcal{N}(0, 0.3^2)$ - and the $\mathcal{N}(0, 0.1^2)$ -distribution. Validation data were generated from a $\text{Bin}(m_i, p_i)$ -distribution with $m_i = 25$, $i = 1, \dots, 800$.

For each data set, MCMC was run for $M = 30\,000$ iterations after a burn-in of 10 000 draws, where the first 5000 draws were from the unrestricted model. Results are summarized in Table 2, where averages of the posterior estimates of the regression effects, averages of the posterior inclusion probabilities for each regressor as well as for the respective random intercept parameter θ over the 50 data sets are shown.

Table 2 Simulation II. Averaged posterior means (Av. Pm.), averaged estimated posterior inclusion probabilities (Av. $\hat{p}_{*,j}$), nominal coverage of (model averaged) 95%-HPD intervals (Cov.) as well as number of models (N_{mod}) where a regressor is included in the final model over 50 data sets. Results for non-zero effects are given in bold.

Model	Covariate	True value	Av. Pm.	Av. $\hat{p}_{*,j}$	Cov.	N_{mod}
Logit	α_0	2.2	2.201	—	0.96	—
	α_1	−0.3	−0.296	1.000	0.96	50
	α_2	0	0.001	0.083	1.00	2
	α_3	−0.3	−0.304	1.000	0.94	50
	α_4	0	0.000	0.039	1.00	0
	θ_α	0.3	0.299	1.000	0.98	50
Poisson	β_0	0.75	0.750	—	0.94	—
	β_1	0.1	0.100	1.000	0.98	50
	β_2	0.1	0.101	1.000	0.94	50
	β_3	0	0.000	0.025	1.00	1
	β_4	0	0.000	0.009	1.00	0
	θ_β	0.1	0.102	1.000	0.96	50

Source: Authors' own.

Additionally, we report the nominal coverage of 95%-HPD intervals (in the 50 data sets) and the number of data sets where a regression effect is classified as non-zero. In both the logit and the Poisson part, all non-zero regression effects are classified as non-zero in all data sets. Only two misclassifications occur for zero effects in the logit and only one in the Poisson model.

The respective random intercept parameter θ is correctly identified as non-zero in both the Poisson and the logit model in all data sets. Figure 3 (a)–(b) illustrates the variation of posterior means of the random intercepts as well as of the random intercept variances for both the logit and the Poisson model. Averages of the posterior means agree well with the true parameter values in both models, but we observe considerably more variation in the logit than in the Poisson model.

Results of a sensitivity study, where different slab variances are compared (see Section B.2 of the electronic supplement), indicate almost no influence of the choice of the slab variance. As suggested by a referee, we investigate in Section B.3 of the supplement, how parameter estimation is affected if validation data is provided only for a subset of clusters and no validation information is available for the remaining clusters. We find only a negligible effect on posterior mean estimates and model selection performance, whereas the coverage of 95%-HPD intervals for θ_α as well as for the intercepts α_0 and β_0 suffers somehow from this lack of information; see Table B5 of the electronic supplement for detailed results.

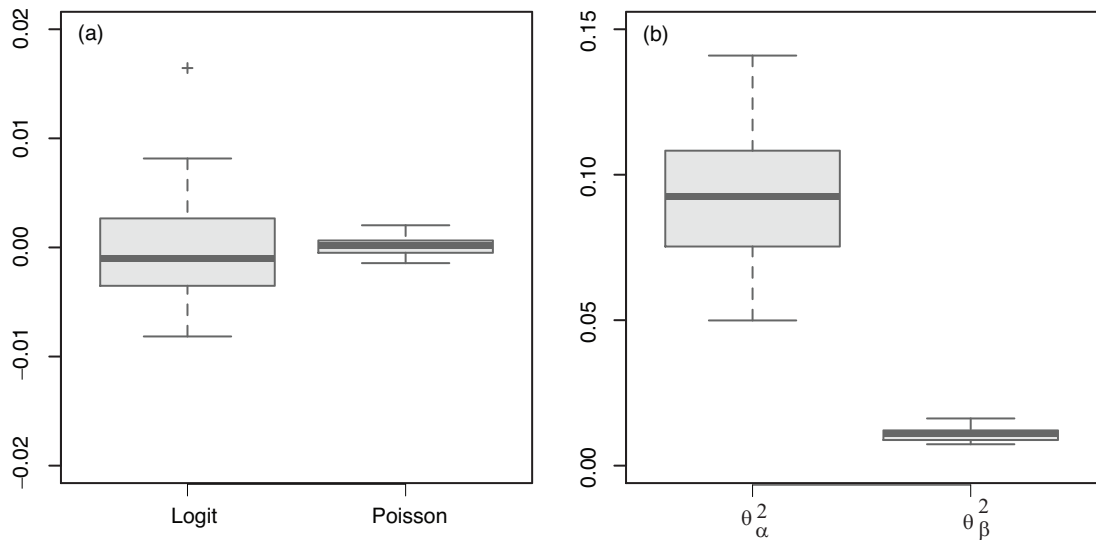


Figure 3 Simulation II. Posterior mean estimates of random intercepts (a) and random intercept variances (b) over 50 simulated data sets.

Source: Authors' own.

5 Application to cervical cancer data

We re-analyse a data set analysed previously in Whittemore and Gong (1991) and Powers *et al.* (2010), which contains the number of cervical cancer deaths for four age groups in four different countries as well as the corresponding woman-years at risk (given in Table 3). Additionally, validation data were collected that give information on how likely physicians from different countries are to identify and correctly report a true cervical cancer death (see Table 4). In that study, a sample of physicians in each country completed a death certificate for one specific patient who had died of cervical cancer and the number of correct death certificates in each country was recorded. Thus, validation data are available on country level but provide no information on the reporting probability specific for age. As a further drawback, with only one case of cervical cancer being assessed, the validation data can capture only variability of the reporting probability across doctors. In our analysis, we will follow Whittemore and Gong (1991) and assume that the validation data are representative for the reporting process, which implies that there is no heterogeneity of reporting across patients.

Whittemore and Gong (1991) consider a Poisson model for risk of death due to cervical cancer and estimate the reporting probabilities by the relative frequencies from the validation data. They consider main effects for age and country with baseline categories age 25–34 and England in the risk model and only country effects with baseline England in the reporting model. Likelihood ratio tests, performed to investigate whether risk and reporting probability differ between countries, suggest country-specific effects on risk of cervical cancer as well as on the reporting probability. Powers *et al.* (2010) analyse the same data but, as noted in Section 3.1,

Table 3 Cervical cancer data. Number of cervical cancer deaths (Y) and woman-years at risk (E, given in thousands of woman-years at risk) for different age categories and European countries between 1969 and 1973.

Country	25–34		35–44		45–54		55–64	
	Y	E	Y	E	Y	E	Y	E
England	192	15399	860	14268	2762	15450	3035	15142
Belgium	8	2328	81	2557	242	2268	268	2253
France	96	15324	477	16186	998	14432	1117	13201
Italy	45	19115	255	18811	621	16234	839	15246

Source: Whittemore and Gong, 1991.**Table 4** Cervical cancer data. Size of validation sample m and number of correct death certificates v .

	England	Belgium	France	Italy
m	50	50	53	50
v	50	43	38	41

Source: Whittemore and Gong, 1991.

in an over-dispersed Póit model. Their results suggest to include all country effects in the reporting model, but only age effects and an effect for Italy in the risk model.

In our analysis, we also specify country effects in the reporting model and country as well as age effects in the Poisson model but additionally include interaction effects. Moreover, we choose a different baseline category as all death certificates are correctly coded for England, and as a consequence, the likelihood would be monotone in the direction of the intercept. Though the posterior is regularized by using a proper prior distribution (see Wagner and Duller, 2012), the prior is influential, which is particularly undesirable for the intercept. We therefore choose Belgium as the baseline category in the reporting model and the lowest age category in Belgium as the baseline category in the Poisson model.

For the parameters of the prior distributions we choose the same values as in the simulation studies, i.e., $m_{\alpha,0} = m_{\beta,0} = 0$, $M_{\alpha,0} = M_{\beta,0} = 100$, the hyper-parameters of slab variances are set to $v = 5$ and $Q = 20$ and the parameters of the hyper-prior for the mixture weights ω_{α} and ω_{β} to $a_{\alpha,0} = b_{\alpha,0} = a_{\beta,0} = b_{\beta,0} = 1$. MCMC was run for 300 000 iterations with thinning of 10 (i.e., 30 000 iterations are retained), after a burn-in of 100 000 draws. The first 20 000 iterations of the burn-in phase were drawn from the unrestricted model.

Table 5 reports the estimated posterior inclusion probabilities, the model averaged posterior means and 95%-HPD intervals for the Poisson risk model as well as for the logit reporting model. In the Poisson model, all main effects of age and country except for France are selected in the median probability model. Compared to Belgium, death risk is higher in England but lower in Italy and death rates increase with age. Three interaction effects are included: The estimated risk of women of age group 55–64 is larger in Italy and that of age groups 45–54 and 55–64 is smaller in France. Though the inclusion probability of the interaction effect for age group 55–64 in France is very close to 0.5, indicating considerable uncertainty on the inclusion of this effect,

Table 5 Cervical cancer data. Model averaged posterior means with 95%-HPD intervals, ratios of HPD lengths with (l_{BMA}) and without (l_*) model selection and estimated posterior inclusion probabilities $\hat{p}_{\beta,j}$ and $\hat{p}_{\alpha,j}$.

		Covariate	Post. mean	95%-HPD	l_{BMA}/l_*	$\hat{p}_{\alpha,j}$
Poisson	β_0	Intercept (Belgium; 25–34)	−4.815	(−4.981, −4.653)	0.312	—
	β_1	England	0.422	(0.279, 0.555)	0.254	1.000
	β_2	France	−0.022	(−0.318, 0.001)	0.273	0.153
	β_3	Italy	−0.935	(−1.098, −0.731)	0.304	1.000
	β_4	35–44	1.604	(1.477, 1.733)	0.230	1.000
	β_5	45–54	2.684	(2.548, 2.816)	0.251	1.000
	β_6	55–64	2.812	(2.651, 3.020)	0.349	1.000
	β_7	England 35–44	−0.001	(−0.098, 0.002)	0.086	0.078
	β_8	England 45–54	−0.001	(0.000, 0.073)	0.066	0.067
	β_9	England 55–64	−0.015	(−0.258, 0.000)	0.236	0.153
	β_{10}	France 35–44	0.026	(0.000, 0.282)	0.236	0.230
	β_{11}	France 45–54	−0.174	(−0.369, 0.000)	0.326	0.779
	β_{12}	France 55–64	−0.104	(−0.368, 0.000)	0.325	0.504
	β_{13}	Italy 35–44	0.006	(−0.008, 0.104)	0.090	0.082
	β_{14}	Italy 45–54	−0.007	(−0.168, 0.007)	0.147	0.093
	β_{15}	Italy 55–64	0.220	(0.000, 0.347)	0.291	0.896
Logit	α_0	Intercept (Belgium)	1.815	(1.084, 2.580)	0.942	—
	α_1	England	3.193	(0.893, 5.797)	0.934	0.994
	α_2	France	−0.851	(−1.711, 0.000)	0.887	0.846
	α_3	Italy	−0.181	(−1.292, 0.454)	0.817	0.453

Source: Authors' own.

the 95%-HPD interval of (−0.368, 0.000) also indicates that the effect for France in this age group might be modified and hence lower than considering only main effects.

In the reporting probability model, only the effect for Italy is not selected based on a posterior inclusion probability above 0.5. Model averaged posterior means for the reporting probabilities are highest in England ($\hat{p}_E = 0.99$), followed by Belgium ($\hat{p}_B = 0.86$) and Italy ($\hat{p}_I = 0.84$) and are lowest for France ($\hat{p}_F = 0.72$).

To illustrate the effect of the spike and slab prior distribution, we compare posterior means and 95%-HPD intervals of the regression effects to those of the unrestricted model, i.e., under a flat prior, in Figure 4. Particularly for the parameters of the Poisson model, there is pronounced shrinkage of small effects to zero resulting in considerably smaller HPD intervals. This gain in precision due to sparse modelling is also evident from the length ratio of 95%-HPD intervals under the spike and slab prior (l_{BMA}) and under the flat prior (l_*), which is reported in Table 5.

The consequences of underreporting are illustrated in Figure 5 which compares posterior mean estimates of country- and age-specific death risk ignoring and accounting for underreporting. Inference using only the observed counts is biased when reporting errors are not corrected for and as a consequence, risks are clearly underestimated in all countries except England, where no underreporting occurs. Risks differ between the countries and these differences seem to be even larger without accounting for country-specific underreporting.

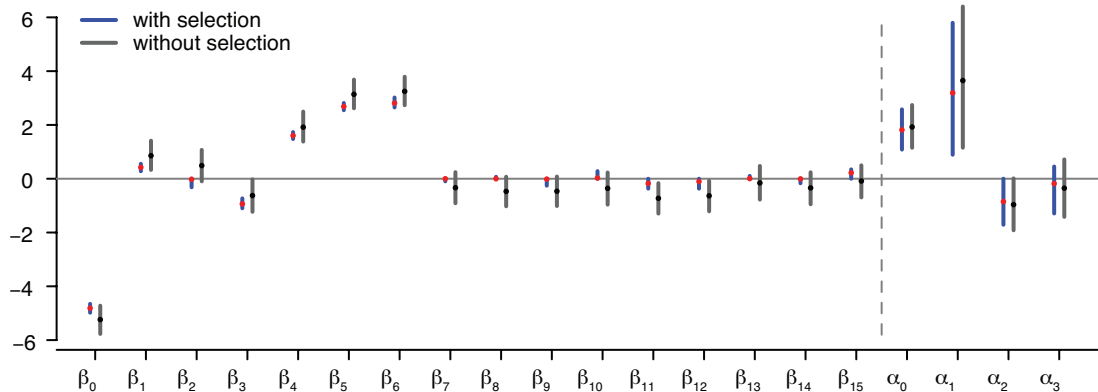


Figure 4 Cervical cancer data. Model averaged posterior means and 95%-HPD intervals for regression effects under the spike and slab prior compared to the unrestricted model.

Source: Authors' own.

As mentioned earlier, differing from previous analyses with England as the reference country, we choose Belgium as the reference country. Figure 6 compares the autocorrelations of the MCMC draws for the effects in the logit model for both reference categories. Obviously, autocorrelations are considerably lower for all country effects except for England and hence, sampling efficiency is much higher with Belgium as the reference category. More details on the sampling efficiency for both reference categories are provided in Section C.1 of the electronic supplement to this paper.

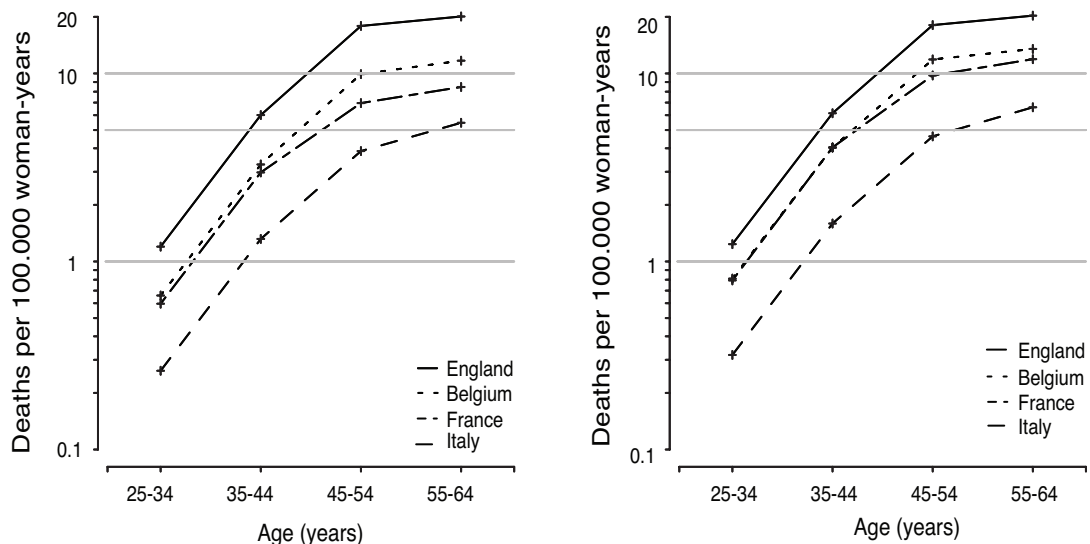


Figure 5 Cervical cancer data. Model averaged country- and age-specific death risk due to cervical cancer ignoring (left) and accounting for underreporting (right).

Source: Authors' own.

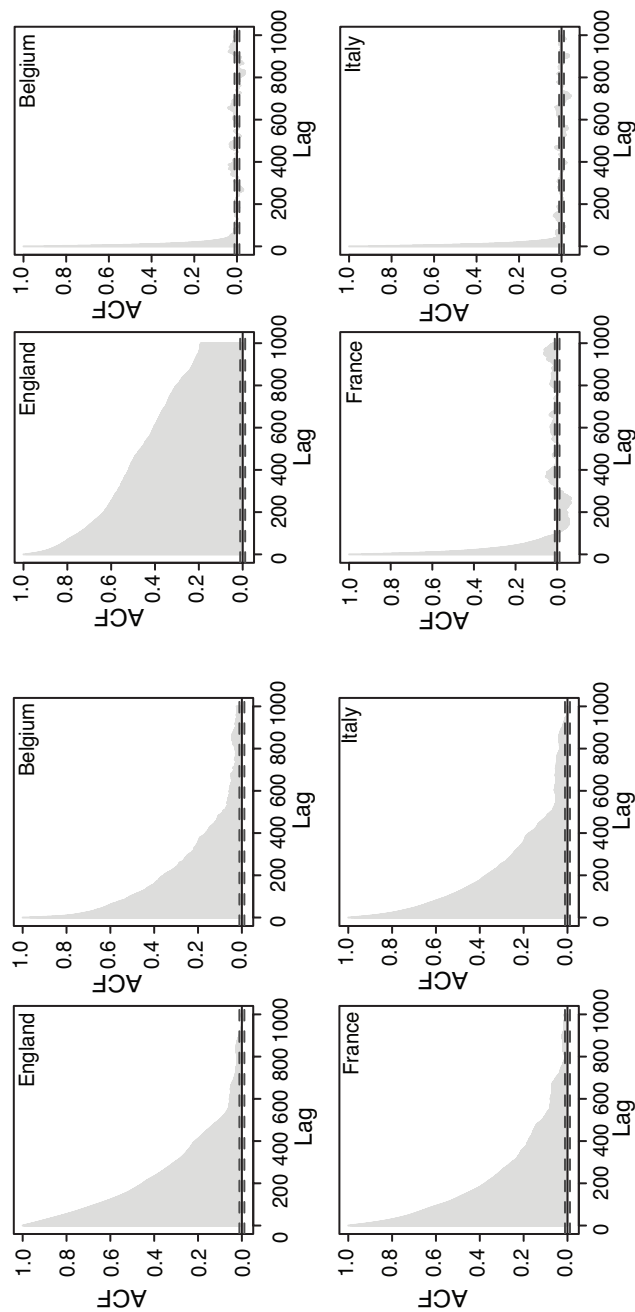


Figure 6 Cervical cancer data. Comparison of autocorrelations in MCMC draws in the logit model for baseline category $\alpha_0 = \text{England}$ (left) and $\alpha_0 = \text{Belgium}$ (right).
Source: Authors' own.

Finally, to assess the adequacy of the selected model, we perform posterior predictive model checks on the full model and the model with covariates x_1, x_3 – x_6, x_{11}, x_{12} and x_{15} in the Poisson and w_1 and w_2 in the logit sub-model, i.e., those covariates selected based on a posterior inclusion probability larger than 0.5. Parameter estimates and 95%-HPD intervals for this model are given in the electronic supplement C.2.

Following Gelman *et al.* (2000), discrete residuals were computed from the MCMC output $(\beta_0^{(m)}, \boldsymbol{\beta}^{(m)}, \alpha_0^{(m)}, \boldsymbol{\alpha}^{(m)})$, $m = 1, \dots, M$ as

$$r_i^{(m)} = y_i - E_i \lambda_i^{(m)} p_i^{(m)}, \quad i = 1, \dots, I,$$

with $\lambda_i^{(m)} = \exp(\beta_0^{(m)} + \mathbf{x}_i \boldsymbol{\beta}^{(m)})$ and $p_i^{(m)} = \text{logit}^{-1}(\alpha_0^{(m)} + \mathbf{w}_i \boldsymbol{\alpha}^{(m)})$ and for each of these draws, a set of replicated data $\tilde{y}_i$ was generated from the posterior predictive distribution.

From Figure 7, which compares 100 plots of the residuals to the 95% predictive bounds for the unrestricted (left) and the selected model (right), we conclude that the overall model fit is reasonably well, even slightly better for the selected model.

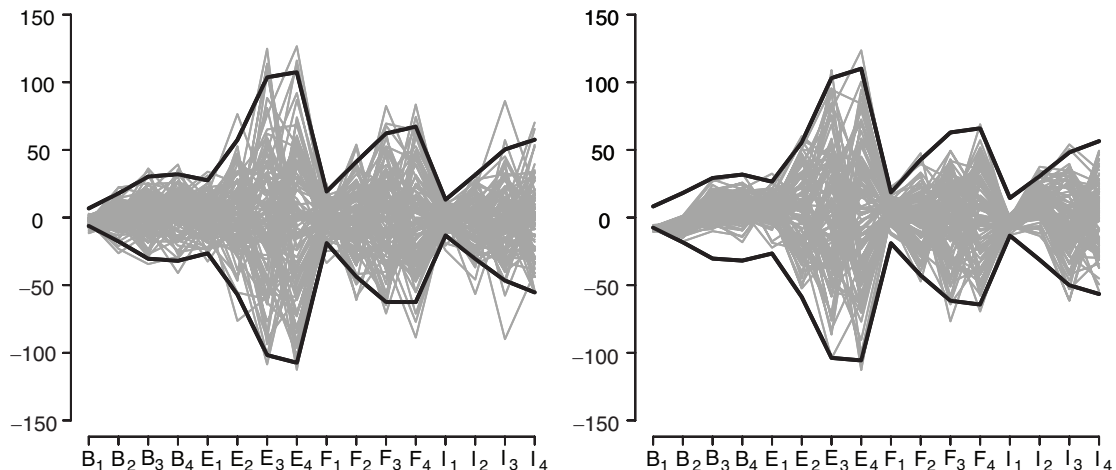


Figure 7 Posterior predictive model checking for the cervical cancer data. 100 plots of residuals and 95% predictive bounds for the unrestricted (left) and the selected model (right).

Source: Authors' own.

6 Concluding remarks

In this paper, we developed a Bayesian framework for variable selection in regression modelling of underreported count data, where both the data generating process of the true number of events as well as the reporting process is allowed to depend on covariates. For posterior inference we propose a convenient MCMC sampling scheme that relies on data augmentation and involves only Gibbs sampling steps. Augmenting

the observed data with the unreported counts has the advantage that the model can be factorized into a Poisson and a binomial logit model and hence, existing auxiliary mixture sampling methods for both sub-models can be used. Thus, extensions to more general model specifications, e.g., including random effects or spatial effects in the linear predictor, are straightforward.

Though we considered the case where an additional validation sample is available, the MCMC scheme can also be used, when only count data subject to underreporting are available. However, in this case, identification of the model parameters has to be guaranteed either by imposing restrictions or specifying (more) informative prior distributions.

For each part of the model, various other choices are possible: For the reporting process, obvious choices would be a probit model, where the latent utility follows a normal distribution or a robit model (Liu, 2004), where the latent utility follows a t -distribution. Alternatives for the count data would be a negative binomial or a Conway-Maxwell-Poisson distribution. Though our implementation of Bayesian variable selection relies on the representation of the likelihood of each sub-model as an auxiliary Gaussian model, this is not necessary when spike distributions are specified as continuous distributions with very small variance as in George and McCulloch (1997) and Ishwaran and Rao (2005). With a continuous spike the indicator variables can be drawn conditional on the regression effects and hence, marginalizing over the regression effects is not required.

Finally, the models considered in this paper can easily be extended to count data subject to both over- and underreporting or even more general misclassification.

Acknowledgements

We thank two anonymous referees for their helpful comments. We are also grateful to Agnes Fussl for providing the R code of the data augmentation algorithm in the binomial logit model.

Appendix A Variable selection for Gaussian regression models

Here we give details on MCMC estimation in a Gaussian regression model with spike and slab priors on the regression coefficients. The MCMC scheme is described for the auxiliary Gaussian regression model resulting for the logit sub-model,

$$y_i^* = \alpha_0 + \mathbf{w}_i \boldsymbol{\alpha} + \varepsilon_{r_i}, \quad \varepsilon_{r_i} \sim \mathcal{N}(0, s_{r_i}^2).$$

We will denote by $\mathbf{y}^*$ the vector of all latent variables, by $\mathbf{W}$ the corresponding design matrix with rows $\mathbf{w}_i$, by $\mathbf{r}$ the vector of indicators for the normal mixture approximation and by $\boldsymbol{\Sigma} = \text{diag}(s_{r_1}^2, \dots, s_{r_n}^2)$ the variance-covariance matrix of the error terms. For a given vector of indicator variables $\boldsymbol{\delta}_\alpha$, the corresponding design matrix will be denoted by $\mathbf{W}^{\boldsymbol{\delta}_\alpha}$ and the regression coefficients by $\boldsymbol{\alpha}^{\boldsymbol{\delta}_\alpha}$.

MCMC involves the following sampling steps:

1. Sample ω_α from the posterior $\omega_\alpha \sim \mathcal{B}(a_{\alpha,0} + \sum \delta_{\alpha,j}, b_{\alpha,0} + d_\alpha - \sum \delta_{\alpha,j})$.
2. Sample the vector of indicators δ_α , the regression coefficients $(\alpha_0, \boldsymbol{\alpha})$ and the scale parameters $\boldsymbol{\psi}_\alpha = (\psi_{\alpha,1}, \dots, \psi_{\alpha,d_\alpha})$ of the slab component:
 - i. Sample each $\delta_{\alpha,j}$ from the conditional posterior $p(\delta_{\alpha,j} | \delta_{\alpha,\setminus j}, \mathbf{y}^*, \omega_\alpha, \boldsymbol{\Upsilon}_\alpha)$, where $\boldsymbol{\Upsilon}_\alpha = (\mathbf{r}, \boldsymbol{\psi}_\alpha)$ and $\delta_{\alpha,\setminus j}$ denotes all elements of δ_α except for $\delta_{\alpha,j}$. The conditional posterior can be computed as

$$p(\delta_{\alpha,j} = 1 | \delta_{\alpha,\setminus j}, \mathbf{y}^*, \omega_\alpha, \boldsymbol{\Upsilon}_\alpha) \propto p(\mathbf{y}^* | \delta_{\alpha,j} = 1, \delta_{\alpha,\setminus j}, \boldsymbol{\Upsilon}_\alpha) \omega_\alpha = \frac{1}{1 + \frac{1-\omega_\alpha}{\omega_\alpha} M_j},$$

where $M_j = \frac{p(\mathbf{y}^* | \delta_{\alpha,j} = 0, \delta_{\alpha,\setminus j}, \boldsymbol{\Upsilon}_\alpha)}{p(\mathbf{y}^* | \delta_{\alpha,j} = 1, \delta_{\alpha,\setminus j}, \boldsymbol{\Upsilon}_\alpha)}$.

Nominator as well as denominator of M_j are conditional marginal likelihoods of a linear regression model with conjugate priors, which are available in closed form, see equation (A.1) below.

- ii. For $\delta_{\alpha,j} = 0$, set $\alpha_j = 0$. Sample $\boldsymbol{\xi}_\alpha = (\alpha_0, \boldsymbol{\alpha}^{\delta_\alpha})$ from the conditional posterior of the reduced regression model

$$\mathbf{y}^* = \mathbf{Z}\boldsymbol{\xi}_\alpha + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}),$$

where $\mathbf{Z} = [\mathbf{1}, \mathbf{W}^{\delta_\alpha}]$. The conditional posterior $p(\boldsymbol{\xi}_\alpha | \mathbf{y}^*, \delta_\alpha, \boldsymbol{\Upsilon}_\alpha)$ is the normal distribution, $\mathcal{N}(\mathbf{a}, \mathbf{A})$, with moments

$$\mathbf{A} = (\mathbf{A}_0^{-1} + \mathbf{Z}'\boldsymbol{\Sigma}^{-1}\mathbf{Z})^{-1}, \quad \mathbf{a} = \mathbf{A}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}\mathbf{y}^*,$$

where $\mathbf{A}_0$ is a diagonal matrix with elements $M_{\alpha,0}$ and the scale parameters $\psi_{\alpha,j}$, for those j where the corresponding indicator $\delta_{\alpha,j}$ takes the value 1. The marginal likelihood in this model is given as

$$p(\mathbf{y}^* | \delta_\alpha, \boldsymbol{\Upsilon}_\alpha) = \frac{1}{(2\pi)^{I/2}} \frac{|\mathbf{A}|^{1/2}}{|\mathbf{A}_0|^{1/2}} \exp\left(-\frac{1}{2}((\mathbf{y}^*)'\boldsymbol{\Sigma}^{-1}\mathbf{y}^* - \mathbf{a}'\mathbf{A}^{-1}\mathbf{a})\right). \quad (\text{A.1})$$

- iii. Sample the scale parameter $\psi_{\alpha,j}$ from the posterior $p(\psi_{\alpha,j} | \delta_{\alpha,j}, \alpha_j)$ for $j = 1, \dots, d_\alpha$. For $\delta_{\alpha,j} = 1$, the conditional posterior is given by

$$\psi_{\alpha,j} | \delta_{\alpha,j}, \alpha_j \sim \mathcal{G}^{-1}\left(v + \frac{1}{2}, \mathcal{Q} + \frac{\alpha_j^2}{2}\right),$$

and otherwise, $\psi_{\alpha,j} \sim \mathcal{G}^{-1}(v, \mathcal{Q})$.

The more general random intercept model requires additional steps to sample the random intercepts and the inclusion probability π_α . The indicator γ_α , the coefficient θ_α and the variance parameter $\psi_{\alpha, d_{\alpha+1}}$ can be sampled jointly with the corresponding parameters in step 2. Full details are given in Wagner and Duller (2012).

For the Poisson sub-model, data augmentation leads to a Gaussian regression model with $I + \sum_{i=1}^I I\{n_i > 0\}$ auxiliary observations. Usually, all augmented counts n_i will be larger than zero and hence, we deal with a linear regression model for $2I$ auxiliary observations $-\log \tau_{i,k}^* - \log E_i$, $k = 1, 2$. Posterior inference for this model involves the same sampling steps as for the logit model.

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